Gursimran Panesar

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Education

Carnegie Mellon University — Aug. 2026
Bachelor of Science in Mathematics (Computer Science & Mathematics)

Experience

Software Engineer — Aug. 2026 – Present
Doordash
- New Verticals: City Fulfillment Networks Team

AI Safety Researcher — Jan. 2026 – Present
Carnegie Mellon University
- First author on research paper targeting ICLR/ICML '27 on cost-efficient long-context safety monitoring for LLMs
- Post-trained and fine-tuned small models through autoresearch to match frontier-model monitoring performance
- Designed new safety benchmarks and evals, measuring monitoring accuracy and cost vs. monolithic SoTA models

Software Engineering Intern — May 2026 – Aug. 2026
Meta
- Migrated 6 Instagram endpoints from Django to Hack, decreasing infrastructure latency for 100M–1B+ users
- Generated $12M in annual infrastructure savings through backend migration resource utilization improvements
- Optimized ranking for follow recommendations to increase follows/reciprocal follows by 2% for daily active users
- Created and integrated automated agentic loops, reducing required human intervention by 70% on migrations

Software Engineering Intern — May 2025 – Aug. 2025
Amazon Web Services (AWS)
- Integrated 10+ diagnostic capabilities into CloudWatch console to reduce context switching during investigation
- Built an MCP server with AI retrieval and tool orchestration to accelerate customer-directed investigations by 60%
- Reduced navigation away from core console by 10%, improving efficiency by keeping users within primary workflow

Software Engineering Intern, AI & Automation — May 2024 – Aug. 2024
Bank of New York (BNY)
- Implemented large language model RAG using PyTorch to read legal loan documents, saving 300hrs per quarter
- Deployed Azure character recognition model for W9 forms to increase processing speed into SQL database by 97%
- Researched AI use-cases and presented 5-year plan to executives for safe implementation of AI/ML technologies

AI Robotics Researcher — Dec. 2023 – Aug. 2024
Carnegie Mellon AirLab
- Programmed critical user mapping features for drone path planning to elevate experience and efficiency using C++
- Created Node.js dashboard for 100 drones to monitor 50 health indicators transmitting from vehicle to base station

Project Experience

AI Sycophancy Evaluation and Mitigation
- Researched sycophancy and mitigation methods in Gemma-4 SoTA model under CMU Prof. Aran Nayebi
- Evaluated and presented mitigation post-training methods Supervised Fine Tuning, Low-Rank Adaptation, Direct Preference Optimization, Reinforcement Learning (with Human Feedback)

Microsoft Tiny Troupe Agentic AI
- Collaborated with Microsoft Research team in evaluating Tiny Troupe AI multiagent persona simulations
- Extended Tiny Troupe with behavioral economics experiments to improve ability to replicate human behavior

ML x Twitter Stock Market
- Developed logistic regression model using Numpy and Pandas to determine sentiment and predict stock trajectory
- Achieved 93% accuracy rate in predicting short term trends of stock prices based on dataset of 184k tweets

Skills & Certifications

Certifications: Amazon AWS Cloud Practitioner, Microsoft Azure AI Fundamentals, IBM Watson AI Fundamentals
Programming: Python, C, Standard ML, JavaScript, Java, Go, C++, SQL, ROS, Unix/Linux
Machine Learning: PyTorch, TensorFlow, Sklearn, Numpy, Pandas, Jupyter, LLMs, RAG, Model Context Protocol
Coursework (Grad/PhD): Generative AI, Deep Learning, Natural Language Processing, Machine Learning